Definitions

Matrix:

=> A rectangular array of $m \times n$ numbers (real or complex) in the form of m horizontal lines (called rows) and n verticle lines (called columns), is called a matrix of order $m \times n$, written as $m \times n$ matrix.

=> Such an array is enclosed in [ ] or ( ).

=> An $m \times n$ matrix is usually written as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$A = [a_{ij}]_{m \times n}$

=> The number $a_{11}, a_{12}, \ldots$ etc, are known as elements of the matrix A, where a_{ij} belongs to the i^{th} row and j^{th} column and is called the $(i, j)^{th}$ element of the matrix $A = [a_{ij}]$.

Types of matrices:

Row matrix: (Also called horizontal)

=> Any matrix which possesses one row and n columns is said to be a row matrix. $n > m$

$A = [a_{11} \ldots \ldots a_{1n}]_{1 \times n}$

e.g $B = [1 \quad -3 \quad 17]$

Column matrix: (Also called vertical)

=> Any matrix which possesses m rows and one column is called column matrix. $m > n$

$$A = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}_{m \times 1}$$

e.g $$Q = \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}$$

Zero matrix or NULL matrix:

=> A matrix that has all elements are zero is called zero or null matrix.

=> it can be of any order

$A = [a_{ij}]_{m \times n}$ where $a_{ij} = 0$

e.g

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Singleton matrix:

=> A matrix which has only one element. i.e ~~one row and one column~~, it is called singleton matrix.

=> Thus, $A = [a_{ij}]_{m \times n}$ is a singleton matrix if $m = n = 1$

e.g

[2], [3], [a], [ ] etc

=> Any real or complex number can be an entry in the singleton matrix.

Square matrix:

=> A matrix in which the no of rows and no of columns are equal, then it is called square matrix.

=> Thus, $A = [a_{ij}]_{m \times n}$ is a square matrix if $m = n$

e.g $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is a square matrix of order 3x3.

Trace of a matrix:

=> The sum of the diagonal elements in a square matrix A is called the trace of matrix A.

=> it is denoted by tr(A).

$tr(A) = \sum_{i=1}^{n} a_{ij} = a_{11} + a_{22} + \dots + a_{mn}$.

i) $tr(\lambda A) = \lambda tr(A)$ ii) $tr(A+B) = tr(A) + tr(B)$ iii) $tr(AB) = tr(BA)$

Diagonal matrix:

=> A square matrix where all the non-diagonal elements are zero except for those in the diagonal from the top left corner to the bottom right corner.

=> Thus, a square matrix $A = [a_{ij}]$ is a diagonal matrix if $a_{ij} = 0$, when $i \neq j$.

e.g

$P = [9]$

$Q = \begin{bmatrix} 9 & 0 \\ 0 & 13 \end{bmatrix}$, $R = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 13 & 0 \\ 0 & 0 & -2 \end{bmatrix}$

Note:

=> A diagonal matrix is always a square matrix.

=> The diagonal elements are characterized by this form.

$A = [a_{ij}]$ when $i \neq j$

Scalar matrix:

=> A matrix in which all the diagonal elements of a diagonal matrix are same or equal. it is called a scalar matrix.

=> $A = [a_{ij}]_{m \times m}$ is a Scalar matrix if $a_{ij} = \begin{cases} 0 & i \neq j \\ k, & i = j \end{cases}$ where k is a constant.

e.g

$$A = \begin{bmatrix} -7 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -7 \end{bmatrix}, \quad P = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$Q = \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & \sqrt{5} \end{bmatrix}$$

Unit matrix or identity matrix:

=> A scalar matrix whose diagonal elements are all 1, is called unit matrix or identity matrix.

=> It is denoted by I_n.

=> Thus, a square matrix $A = [a_{ij}]_{m \times n}$ is identity matrix if:

$$a_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

imp points:

- All identity matrices are scalar matrices.
- All scalar matrices are diagonal matrices.
- All diagonal matrices are square matrices.
- The converse of these statements is not true.

Equal matrices:

=> Two matrices A and B are said to be equal:

- if they are of the same order
- their corresponding elements are equal.

=> Two matrices $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{r \times s}$ are equal if:

a) $m = r$, i.e. the number of rows in A is equal the number of rows in B.

b) The number of columns in A are equal to the number of columns in B. $n = s$.

c) $a_{ij} = b_{ij}$, for $i = 1, 2, \ldots, n$ and $j = 1, 2, \ldots, n$, i.e. the corresponding elements are equal.

$A = [a_{ij}]_{m \times n}$ & $B = [b_{ij}]_{r \times s}$ where $a_{ij} = b_{ij}$, $m = r$, $n = s$

e.g $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ are not equal because their orders are not the same.

But if

$A = \begin{bmatrix} 1 & 6 & 3 \\ 5 & 2 & 1 \end{bmatrix}$ and $\begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$

are equal matrices then:

$a_1 = 1$, $a_2 = 6$, $a_3 = 3$, $b_1 = 5$, $b_2 = 2$, $b_3 = 1$

Triangular matrix:

=> A square matrix is said to be triangular matrix if the elements above or below the principal diagonal are zero.

Upper Triangular

=> A square matrix $[a_{ij}]$ is called an upper triangular matrix, if $[a_{ij}] = 0$ when $i > j$.

e.g $\begin{bmatrix} 3 & 1 & 2 \\ 0 & 4 & 3 \\ 0 & 0 & 6 \end{bmatrix}$ of 3×3.

Lower Triangular:

=> A square matrix is called lower triangular matrix, if $a_{ij} = 0$ where $i < j$

e.g $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 2 \end{bmatrix}$ of order 3×3.

Singular matrix:

=> A matrix A is said to be a singular matrix if its determinant is zero

$|A| = 0$

Non- Singular matrix:

=> A matrix is said to be non-singular matrix if its determinant not equal to zero.

$|A| \neq 0$.

Symmetric matrix:

=> A matrix whose transpose is the same as the orignal matrix is called a symmetric matrix.

=> Only a square matrix can be a symmetric matrix.

=> A square matrix $A = [a_{ij}]$ is called symmetric matrix if $a_{ij} = a_{ji}$ for all values.

$$A^t = A$$

e.g.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 2 \end{bmatrix}$$

$$A^t = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 2 \end{bmatrix}$$

$$A = A^t$$

Skew-Symmetric matrix:

=> A square matrix $A = [a_{ij}]$ is a skew-symmetric matrix if $a_{ij} = -a_{ji}$ for all values of i, j
[putting $j = i$] $a_{ii} = 0$

=> In skew symmetric matrix all diagonal elements are zero.

=> A square matrix is skew-symmetric matrix $A^t = -A$

e.g

$$A = \begin{bmatrix} 0 & 2 & 1 \\ -2 & 0 & -3 \\ -1 & 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

Imp points:

=> if A is any square matrix, then $A + A^t$ is a symmetric matrix and $A - A^t$ is a skew symmetric matrix.

=> if AB is symmetric matrices then BA also symmetric matrices.

Proof $A + A^t$ is symmetric:

Let A be a square matrix:

$$(A + A^t)^t = A^t + (A^t)^t$$
$$= A^t + A$$
$$= A + A^t$$

Prove $A-A^t$ is Skew symmetric

$$(A-A^t)^t$$
$$= A^t - (A^t)^t$$
$$= A^t - A$$
$$= -(A-A^t)$$

Theorem:

⇒ Every square matrix A can be decomposed uniquely as the sum of two matrices S and V, where S is symmetric and V is skew-symmetric.

$$S = \frac{1}{2}(A+A^t)$$
$$V = \frac{1}{2}(A-A^t)$$
$$= \frac{1}{2}(A+A^t) + \frac{1}{2}(A-A^t)$$
$$= \frac{1}{2}(B+C)$$

⇒ Addition and difference of two Symmetric matrices results in symmetric matrix.

⇒ if A is a symmetric matrix then A^{-1} is also symmetric.

⇒ when identity matrix is added to a skew symmetric matrix then the resultant matrix is invertible.

Hermitian matrices:

⇒ A square matrix A is said to be Hermitian matrix if:

$$(\bar{A})^t = A$$

⇒ e.g $A = \begin{bmatrix} 1 & 1-i \\ 1+i & 2 \end{bmatrix}$

taking conjugate

$$\bar{A} = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}$$

$$(\bar{A})^t = \begin{bmatrix} 1 & 1-i \\ 1+i & 2 \end{bmatrix} = A$$

∴ So A, is hermitian matrix.

⇒ every diagonal element of a hermitian matrix must be real.

skew-hermitian matrix:

⇒ A square matrix A is said to be skew-hermitian if

$$(\bar{A})^t = -A$$

e.g $A = \begin{bmatrix} 0 & 2-3i \\ -2-3i & 0 \end{bmatrix}$

taking conjugate

$\bar{A} = \begin{bmatrix} 0 & 2+3i \\ -2+3i & 0 \end{bmatrix}$

taking transpose

$(\bar{A})^t = \begin{bmatrix} 0 & -2+3i \\ 2+3i & 0 \end{bmatrix} = -A$

Conjugate matrix:

$\Rightarrow$ Let A be a matrix over the field of complex number. The change each element of A by it's conjugate. The matrix obtained is called conjugate matrix of A, and denoted by $\bar{A}$

e.g. $A = \begin{bmatrix} 1+i & 2-i \\ 3 & i \end{bmatrix}$

$\bar{A} = \begin{bmatrix} 1-i & 2+i \\ 3 & -i \end{bmatrix}$

Idempotent matrix:

$\Rightarrow$ A square matrix A is said to be idempotent matrix if $A \cdot A = A$

$\Rightarrow A^2 = A$

Involutary matrix:

$\Rightarrow$ A square matrix A is said to be involutary if $A \cdot A = I_A$

$A^2 = I$

NILpotent matrix:

$\Rightarrow$ A Square matrix A is said to be nilpotent matrix if $A^p = 0$ (where p is +ive integer)

=> if p is the least +ive integer for which $A^p = O$, Then A is said to be nilpotent of nilpotency index p.

Periodic matrix:

=> A square matrix A is said to be periodic matrix if

$$A^{k+1} = A$$

where k is a +ive integer is called a periodic matrix Will period k:

$$A^2$$

$$\vdots$$

$$A^{k+1} = A$$

e.g $A^3 = A$

$A^{2+1} = A$, $k = 2$

Orthogonal matrix:

=> A square matrix A is said to be Orthognal matrix if its transpose is the same as its inverse, that is

if

$$A^{-1} = A^t$$

or, equivalently, if

$$AA^t = A^tA = I$$

Properties:

- $AA^t = A^tA = I$
- $A^t = A^{-1}$
- Identity matrix is also orthogonal matrix.
- All diagonal matrix is also orthognal matrix if (1, -1)
- $\det(A) = \pm 1$
- if A and B are orthognal matrix then AB and BA will be always orthognal.

Unitary matrix:

A complex square matrix A is called unitary matrix if

$$AA^t = A^tA = I$$

$$AA^t = I \quad \text{---(i)}$$

if we multiply by A^{-1}

$$A^{-1}AA^t = A^{-1}I$$

$$IA^t = A^{-1}I$$

$$\boxed{A^t = A^{-1}}$$

e.g.

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & 1/\sqrt{2} & 0 \\ \frac{-i}{\sqrt{2}} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Properties:

1) Det of A, $|\det(A)| = 1$
2) Each row & column of an unitary matrix is normalized.

transpose of a matrix:

=> The new matrix obtained by interchanging the rows and columns of the original matrix is called as the transpose of the matrix.

=> if $A = [a_{ij}]_{m \times n}$, then $A^t = [a_{ji}]_{n \times m}$.

=> it is denoted by A^t,

Properties:

=> $(A^t)^t = A$

=> $(A \pm B)^t = A^t \pm B^t$

=> $(AB)^t = B^t A^t$

=> $(KA)^t = K(A)^t$

=> $(A_1 A_2 A_3 \ldots, A_{n-1} A_n)^t = A^t_n A^t_{n-1} \ldots$

=> $I^t = I$

=> $tr(A) = t(A^t)$

=> $(A^{-1})^t = (A^t)^{-1}$

=> $\det(A) = \det(A^t)$

Prove that $(KA)^t = KA^t$

Proof:

$= \quad A = (a_{ij}) \Rightarrow A^t = a_{ji}$

$= KA = (K a_{ij}) \Rightarrow (KA)^t = (K a_{ji})$

$= KA^t = K a_{ji}$

$(KA)^t = KA^t$

Prove that $(AB)^t = B^t A^t$

Proof:

$A_{m \times n}$, $B_{n \times p}$

$A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{n \times p}$

$A^t = (a_{ji})_{m \times n}$, $B^t = (b_{ji})_{n \times p}$

$$(AB_{ij})^t = \sum_{k=1}^{n} (a_{ik} b_{kj})^t$$

$$= \sum_{k=1}^{n} (a_{jk} b_{ki})$$

$$= \sum_{k=1}^{n} (b_{ki} a_{jk})$$

$$= ((B^t A^t)_{ij})$$

Prove that $(A+B)^t = A^t + B^t$

Proof:

$A = (a_{ij}) \Rightarrow A^t = a_{ji}$

$B = (b_{ij}) \Rightarrow B^t = a_{ji}$

$A+B = a_{ij} + b_{ij}$

$(A+B)^t = (a_{ij} + b_{ij})^t$

$= a_{ji} + b_{ji}$

$= A^t + B^t$

Prove $(A^t)^t = A$.

Proof:

$A = a_{ij}$

$A^t = (a_{ij})^t = a_{ji}$

$(A^t)^t = (a_{ji})^t = a_{ij} = A$

$\therefore (A^t)^t = A$

Show that AB is invertible if A & B are invertible matrices.

Proof:

$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$

$= AIA^{-1} = AA^{-1}$

$= I$

Prove that $(AB)^{-1} = B^{-1}A^{-1}$

$= (B^{-1}A^{-1})AB = B^{-1}(A^{-1}A)B = B^{-1}IB$

$= B^{-1}B = I$

$(AB)(B^{-1}A^{-1}) = I$ So AB is invertible.

Prove that $(A^{-1})^t = (A^t)^{-1}$

Proof:

$\because |A| = |A^t|$

$\Rightarrow |A^t| \neq 0$

$\Rightarrow A^t$ is also inveotible

$\Rightarrow AA^{-1} = I_n = A^{-1}A$

$\Rightarrow (AA^{-1})^t = (I_n)^t = (A^{-1}A)^t$ $\qquad \because (AB)^t = B^t A^t$

$\Rightarrow (A^{-1})^t A^t = I_n = (A^{-1})^t A^t$

$\Rightarrow (A^t)^{-1} = (A^{-1})^t$

Prove that $AA^{-1} = A^{-1}A = I$

Proof:

$\Rightarrow AB = BA = I$ then B is said to be inverse matrix of A. it is denoted by A^{-1}

$\boxed{AA^{-1} = A^{-1}A = I}$

Prove that $(A^{-1})^{-1} = A$

Let $(A^{-1})^{-1} = B$

$\Rightarrow (A^{-1})(A^{-1})^{-1} = A^{-1}B$

$I = A^{-1}B$

Poemultyply b. A

$AI = (AA^{-1})B$

$AI = IB$

$A = B \Rightarrow \boxed{(A^{-1})^{-1} = A}$

→ **Method 2:**

$(A^{-1})^t = (A^t)^{-1}$

Proof: Let $A^t (A^t)^{-1} = I$ | $AA^{-1} = I$

Taking transpose b/s

$\Rightarrow (A^t (A^t)^{-1})^t = I^t$ | $(AB)^t = B^t A^t$

$\Rightarrow ((A^t)^{-1})^t (A^t)^t = I$

$\Rightarrow ((A^t)^{-1})^t \cdot A = I$

$\Rightarrow ((A^t)^{-1})^t (AA^{-1}) = IA^{-1}$ (Post multiply by A^{-1}

$\Rightarrow ((A^t)^{-1})^t I = IA^{-1}$

$\Rightarrow ((A^t)^{-1})^t = A^{-1}$

$\Rightarrow$ Taking transpose b/s

$\Rightarrow [((A^t)^{-1})^t]^t = (A^{-1})^t$

$\Rightarrow \boxed{(A^t)^{-1} = (A^{-1})^t}$

Q. Prove that $(A^k)^{-1} = (A^{-1})^k$, $k \in N$

Proof: $(A^k)^{-1} (A^k) = I$

$\Rightarrow (A^k)^{-1} A^k A^{-1} = IA^{-1}$

$\Rightarrow (A^k)^{-1} A^{k-1} = A^{-1}$

$\Rightarrow (A^k)^{-1} (A^{k-1}) [A^{-1} . A^{-1} \ldots (k-1) \text{ times}]$

$= A^{-1} [A^{-1} A^{-1} \ldots (k-1) \text{ times}]$

$\Rightarrow (A^k)^{-1} = (A^{-1})^t$

Prove that $|A^{-1}| = |A|^{-1} = \frac{1}{|A|}$

Proof:

$\Rightarrow AA^{-1} = I$

$\Rightarrow |AA^{-1}| = |I|$ $\boxed{A^{-1} \neq \frac{1}{A}}$

$\Rightarrow |A|\,|A^{-1}| = 1$

$\Rightarrow |A^{-1}| = \frac{1}{|A|}$

Prove that $\text{adj}\, A^{-1} = (\text{adj}\, A)^{-1}$

Proof:

$\Rightarrow A\,(\text{adj}\, A) = |A|\, I$

$\Rightarrow \cancel{AA^{-1}}(\text{adj}\, A) = AA^{-1}$ ——— (i)

$\Rightarrow A\,(\text{adj}A)(\text{adj}\, A)^{-1} = |A|\,(\text{adj}A)^{-1}$

$A = |A|\,(\text{adj}\, A)^{-1}$

$(\text{adj}\, A)^{-1} = \frac{A}{|A|}$ ——— (2)

$\text{adj}\, A^{-1} = |A^{-1}|\,(A^{-1})^{-1}$

$= |A|^{-1} A$

$(\text{adj}\, A)^{-1} = \frac{A}{|A|}$

Difference between hermitian and unitary matrix.

- A Hermitian matrix is a self-adjoint matrix

$$A = A^{+}$$

- An unitary matrix matrix is a matrix with its adjoint equals to its inverse.

$$A^{+} = A^{-1}$$

- The inverse and adjoint of a unitary matrix is also unity.

Prove that det(A) = det(At).

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$\det(A) = a_{11}a_{22} - a_{12}a_{21}$

$At = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$

$\det(At) = a_{11}a_{22} - a_{12}a_{21}$

Hence, prove that

L.H.S = R.H.S

Prove that if A^{-1} exist, then $\det(A) \neq 0$.

$\det(AB) = \det(A)\det(B)$

we know that

$A \cdot A^{-1} = I = A^{-1}A$

$AA^{-1} = I$

$\det(AA^{-1}) = \det(I)$

$\det(A)\det(A^{-1}) = 1$

$0 \cdot \det(A^{-1}) \neq 0$

thus $\det(A) \neq 0$

Q. if A is invertible matrix and n is non-negative integer, then show that $(A^n)^{-1} = (A^{-1})^n$.

if A is invertible —— given

mean A^{-1} exist

$A^n = (A \cdot A \ A \ldots n \text{ factor } A)$

$(A^n)^{-1} = (A \cdot A \cdot A \ldots n \text{ factor } A)^{-1}$

$(A^n)^{-1} = A^{-1} A^{-1} A^{-1} \ldots n \text{ factor } A^{-1}$

$(A^n)^{-1} = (A^{-1})^n$

Topic:

Operations on Matrices:

The following operations could be performed on matrices:

- Addition
- subtraction
- multiplication

Addition:

- Let $A, B \in M_{m \times n}$, then addition of matrices is define by:

$$(A+B)_{ij} = (A)_{ij} + (B)_{ij}$$

- order of matrices should be the same.

Properties:

$$A+B = B+A$$

$$A+(B+C) = (A+B)+C$$

Scalar multiplication:

- if A is any matrix and k is any number, the scalar multiple kA is the matrix obtained from A by multiplying each entry of A by k.

If $KA = 0$, Show that either $K=0$ or $A=0$.

Let

$A = [a_{ij}]$

if $KA = 0$ means $Ka_{ij} = 0$ for all i and j.

if $K=0$ there is nothing to do

if $K \neq 0$ then $Ka_{ij} = 0$ implies that $a_{ij} = 0$ for all i and j, that is $A=0$

Properties:

- $K(A+B) = KA + KB$
- $(K+P)A = KA + PA$
- $(KP)A = K(PA)$
- $1A = A$.

Multiplication:

- Let A be an $m \times n$ matrix and B be an $n \times k$ matrix and write $B = [b_1, b_2 ... b_k]$
- The product matrix AB is the $m \times k$ matrix defined as follows:
- $AB = A[b_1, b_2, \ldots, k_b] = [Ab_1, Ab_2 \ldots Ab_k]$

Properties:

- $AB \neq BA$
- $A(BC) = (AB)C$
- $A(B+C) = AB + AC$
- $(B+C)A = BA + CA$
- $\gamma(AB) = (\gamma A)B = A(\gamma B)$
- $I_m A = A I_n = A$
- if $AB = AC$, it is not required that $B = C$
- if $AB = 0$ you cannot conclude $A = 0$ or $B = 0$

$\therefore$ $(m \times n)(n \times p) = (m \times p)$

$$\begin{bmatrix} m \text{ rows} \\ n \text{ columns} \end{bmatrix} \begin{bmatrix} n \text{ rows} \\ p \text{ colums} \end{bmatrix} = \begin{bmatrix} m \text{ rows} \\ p \text{ columns} \end{bmatrix}$$

- Square matrices can be multiplied if they have same size.
- Each column of AB is a combination of the columns of A

Block Matrices:

⇒ A block matrix is a matrix whose elements are themselves matrices which are called submatrices.

⇒ Oftentimes, there are clear patterns in the entries of a large matrix and it might be useful to break that matrix down into smaller chunks based on some partition of its rows and columns.

for example:

$$A = \left[\begin{array}{ccc:ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ \hdashline 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right]$$

$$A = \begin{bmatrix} I_3 & I_3 \\ O_{33} & C \end{bmatrix}, \text{ where } C = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -2 & 3 \end{bmatrix}$$

Multiplication of block mat

Given that:

$$A = \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{array}\right], \quad B = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 3 & 6 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$$A = \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 3 & 4 & 0 \\ \hline 0 & 0 & 2 \end{array}\right], \quad B = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 3 & 6 & 1 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

$AB = ?$

Let $A_{11} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $A_{13} = \begin{bmatrix} 0 & 0 \end{bmatrix}$

$A_{12} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $A_{14} = \begin{bmatrix} 2 \end{bmatrix}$

Let $B_{11} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \end{bmatrix}$

$B_{13} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$

$B_{12} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $B_{14} = [1]$

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{13} & A_{14} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{13} & B_{14} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{13} & A_{11}B_{12} + A_{12}B_{14} \\ A_{13}\cdot B_{11} + A_{14}B_{13} & A_{13}B_{12} + A_{14}B_{14} \end{bmatrix}$$

$$(A_{11})(B_{11}) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times1+2\times4 & 1\times2+2\times3 & 1\times3+2\times6 \\ 3\times1+4\times4 & 3\times2+4\times3 & 3\times3+4\times6 \end{bmatrix}$$

$$= \begin{bmatrix} 1+8 & 2+6 & 3+12 \\ 3+16 & 6+12 & 9+24 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 15 \\ 19 & 18 & 33 \end{bmatrix}$$

$$(A_{12})(A_{13}) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times0 & 1\times0 \\ 0\times0 & 0\times0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(A_{11})(B_{12}) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times1+2\times1 \\ 3\times1+4\times1 \end{bmatrix} = \begin{bmatrix} 1+2 \\ 3+4 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

$$(A_{12})(B_{14}) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} [2]$$

$$= \begin{bmatrix} 1\times 2 \\ 0\times 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$(A_{13})(B_{11}) = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 0\times1+0\times4 & 0\times2+0\times3 & 0\times3+0\times6 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$(A_{14})(B_{13}) = [2] \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2\times0 & 2\times0 & 2\times0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$(A_{13})(A_{12}) = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0\times1+0\times1 \end{bmatrix}$$

$$= [0]$$

$$(A_{14})(B_{14}) = [2]\,[1]$$

$$= [2]$$

$$(A_{12})(B_{13}) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1\times0 & 1\times0 & 1\times0 \\ 0\times0 & 0\times0 & 0\times0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(A_{11}B_{11} + A_{12}A_{13}) = \begin{bmatrix} 9 & 8 & 15 \\ 19 & 18 & 33 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 9+0 & 8+0 & 15+0 \\ 19+0 & 18+0 & 33+0 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 15 \\ 19 & 18 & 33 \end{bmatrix}$$

$$A_{11}B_{12} + A_{12}B_{14} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3+1 \\ 7+0 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$A_{13}B_{11} + A_{14}B_{13} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$A_{13}B_{12} + A_{14}B_{14} = \begin{bmatrix} 0 \end{bmatrix} + \begin{bmatrix} 2 \end{bmatrix} = \begin{bmatrix} 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 9 & 8 & 15 & 4 \\ 19 & 18 & 33 & 7 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Topic #

Q: Define Similar matrices.

⇒ Let A and B be square matrices of same size then $A \sim B$ (Similar) if $\exists$ an invertible matrix P such that

$$A = P^{-1}BP$$

$$\text{or } PA = BP$$

$$PAP^{-1} = BPP^{-1} = B$$

$$PAP^{-1} = B \Rightarrow B \sim A$$

if $A \sim B \Longleftrightarrow B \sim A$

Properties:

if $A \sim B$ are similar, then

- $|A| = |B|$
- $C(A) = C(B)$
- A and B has same eigen value
- A and B has same rank and nulity
- $Tr(A) = Tr(B)$

How to find Similar matrix?

if A is given:

- Characteristic polynomial $A - \lambda I$
  $$|A - \lambda I| = 0$$
- $\lambda_1, \lambda_2, \ldots, \lambda_n$ (Eigen values)

$\Rightarrow x_1, x_2, \dots, x_n$ (Eigen vector)

$\Rightarrow$ "P" is obtained by combining all eigen vector into one matrix

$\Rightarrow$ find P^{-1}

$\Rightarrow$ $P^{-1}AP \Rightarrow B$ (Answer)

B is called similar matrix.

To check:-

B is similar to A or Not.

Question:-

$$A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$ and λ be eigenvalue of A

then

$$A - \lambda I = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 3 \\ -1 & -2-\lambda \end{bmatrix}$$

$$|A| = (2-\lambda)(-2-\lambda) - (-1)(3)$$

$$= -4 - 2\lambda + 2\lambda + \lambda^2 + 3$$

$$= \lambda^2 - 1$$

Eigen values:

$|A - \lambda I| = 0$

$\lambda^2 - 1 = 0$

$\lambda^2 = 1$

$\Rightarrow \lambda = \pm 1$

$\left\{ \begin{array}{l} \lambda = 1 \\ \lambda_2 = -1 \end{array} \right.$ are eigen values.

Eigen vectors:

$|A - \lambda I| V = 0$

$$\begin{bmatrix} 2-\lambda & 3 \\ -1 & -2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\boxed{\text{At } \lambda = 1}$

$$\begin{bmatrix} 2-1 & 3 \\ -1 & -2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x_1 + 3x_2 = 0$ $\qquad -x_1 - 3x_2 = 0$

$x_1 = -3x_2$

Let $x_2 = t$

$\boxed{x_1 = -3t}$

$$V = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -3t \\ t \end{bmatrix} = t \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$\boxed{\text{At } \lambda = -1}$

$|A - \lambda I| v = 0$

$$\begin{bmatrix} 2-\lambda & 3 \\ -1 & -2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2+1 & 3 \\ -1 & -2+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$3x_1 + 3x_2 = 0$ $\qquad$ $-x_1 - x_2 = 0$

$x_1 + x_2 = 0$

$x_1 = -x_2$ $\qquad$ Let $x_2 = t$

$x_1 = -t$

$$V_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$\Rightarrow$ P matrix is:

$$P = \begin{bmatrix} -3 & -1 \\ 1 & 1 \end{bmatrix}$$

$|P| = (-3)(1) - (-1)(1) = -3 + 1 = -2$

$\neq 0$

P is non-singular.

$$P^{-1} = \frac{\text{Adj of } P}{|P|} = \frac{\begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}}{-2}$$

$$P^{-1}AP = \frac{-1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} -3 & -1 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} -6+3 & -2+3 \\ 3+2 & 1+2 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} -3 & 1 \\ 5 & 3 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} -3+1 & 1+1 \\ 3-3 & -1+3 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} -2 & 2 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

To check

$$|B - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(-1-\lambda) - 0 = 0$$

$$-1 - \lambda + \lambda + \lambda^2 = 0$$

$$\lambda^2 - 1 = 0 \Rightarrow \boxed{\lambda = \pm 1}$$

⇒ Eigen value of B matrix are similar to A matrix so

$$A = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \text{ is similar to } B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Example of trace of matrix.

$$tr(A) = \sum_{i=1}^{n} a_{ii}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$ find $tr(AA^t)$?

$$AA^t = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+9 & 4+10+18 & 7+16+27 \\ 4+10+18 & 16+25+36 & 28+40+54 \\ 7+16+27 & 28+40+54 & 49+64+81 \end{bmatrix}$$

$$= \begin{bmatrix} 14 & 22 & 50 \\ 32 & 77 & 122 \\ 50 & 122 & 194 \end{bmatrix}$$

$$(AA^t) = 14 + 77 + 194 = 285$$